

Serving Sentiment:

Identifying Key Themes in McDonald's Customer Reviews

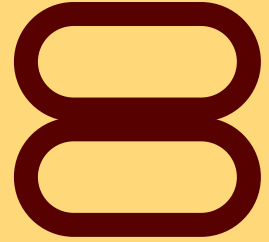
Group 5

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Project Details

- **Motivation:**
 - Only 4% of consumers make a purchase without reading reviews
 - 42% would trust reviews as much as a family member or friend
 - Reviews are important to understand what customers value most in their restaurant experience and can provide insight on what areas require immediate attention to elevate customer experiences at McDonald's
- **Hypotheses:**
 - At least 30% of positive reviews are related to the quality of food
 - At least 30% of negative reviews are related to customer service
- **Research Question:**
 - What are common themes that emerge in positive and negative McDonald's restaurant reviews?
- **Modeling Approach:**
 - Topic Modeling – Latent Dirichlet Allocation (LDA)
- **Goal:**
 - To identify key themes that appear in positive and negative McDonald's reviews

Data Acquisition & Explanation

Column	Description	Potential Responses
reviewer_id	Unique Identifier for each reviewer	1, 227, 33,396
review	Textual context of review	The staff are very friendly and they do their job perfectly
rating	Rating provided by reviewer	1 star, 2 stars, 5 stars
rating_number_only	Cleaned ratings to only include number	1, 2, 5
topic	Number correlating to each of the six different topic categories.	1 - 6
topic probability	Our model's estimated probability that a given review belongs to one of the six predetermined categories	0 - 1
topic label	Content of the six different topic categories	Bad service, wrong orders, rude staff
sentiment	Assigned sentiment of review	positive, negative, neutral

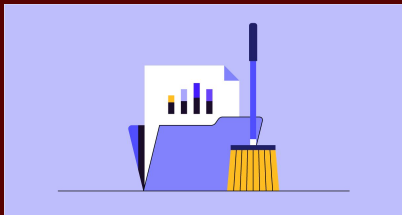
Acquisition: The original dataset was obtained from Kaggle with no licensing or ethical concerns. The file was in a CSV format and contained 33,396 total observations and 10 columns. We focused on the reviews and ratings (text data).

We chose this option simply due to the sheer number of accessible reviews allowing us to focus more of our energy on the analysis and less on scraping data.

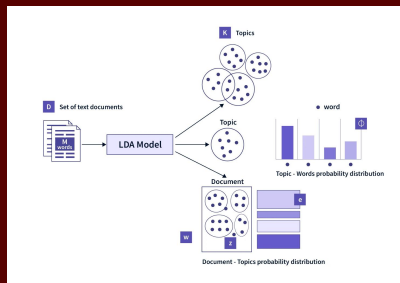
We then cleaned our data set by standardizing responses and correcting variable types. As part of our LDA process, we assigned topics to each review.

This brought us to our final data set that would be utilized in our analysis (seen to the left).

Analysis Plan & Justification



- Lowercase
- Remove punctuation, digits, stopwords
- Lemmatization



- Tokenize reviews
- Document-Term matrix
- Choose k
- LDA model
- Assign topics and compare between different sentiments



- 2 one-proportion z-tests
- Positive reviews & food quality
- Negative reviews & bad service

● Challenges:

- Runtime constraints (large dataset)
- Highest coherence score was for 3 topics

● Diagnostic/validation work:

- Coherence scores were compared across multiple k values
- Perplexity and coherence scores used for evaluation of final model

Tricky Analysis Decision

Selecting the k value for the LDA model

- The k value indicates how many topics the model should extract from the McDonald's reviews
- Chosen based on coherence to emphasize interpretability
- Smaller k values imply broader topics
- k=6 was selected to balance coherence and meaningful results to compare positive and negative reviews
 - Allowed us to distinguish between service, food quality, and timeliness
 - Led us to have a miscellaneous/general topic
 - Goal statement was changed from identifying the top 3 areas most in need of improvement to the top 1–2 areas
 - For hypothesis testing, chose “Overall good price, quality, service” to represent food quality

Coherence scores:

k value: 3 Coherence: 0.5874652089286838

k value: 4 Coherence: 0.533791815339238

k value: 5 Coherence: 0.5282919147370571

k value: 6 Coherence: 0.5260010350832833

k value: 7 Coherence: 0.513929268858304

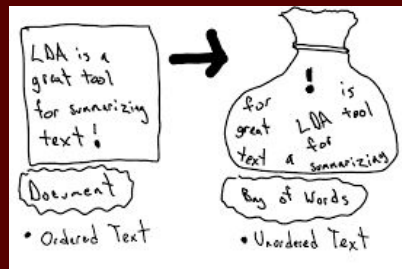
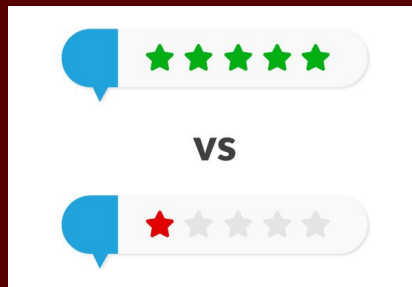
The following output is the top keywords per topic. The LDA model produces 6 topics (k=6), and it learns which words are most commonly associated with each topic. Here are the topics (0–6) and the associated words and weights.

```
{(0, '0.082*order' + 0.065*bad' + 0.041*service' + 0.036*manager' + 0.031*customer' + 0.023*rude' + 0.022*food' + 0.021*mcdonald' + 0.018*employee' + 0.018*wrong'), (1, '0.106*service' + 0.067*food' + 0.060*fast' + 0.057*great' + 0.056*place' + 0.049*clean' + 0.047*staff' + 0.043*nice' + 0.041*friendly' + 0.024*quick'), (2, '0.291*good' + 0.097*excellent' + 0.055*mcdonald' + 0.049*food' + 0.034*attention' + 0.033*service' + 0.024*place' + 0.018*price' + 0.013*breakfast' + 0.013*quality'), (3, '0.067*fry' + 0.035*meal' + 0.032*burger' + 0.026*sandwich' + 0.022*chicken' + 0.021*poor' + 0.020*old' + 0.020*cold' + 0.019*nugget' + 0.019*large'), (4, '0.058*time' + 0.047*order' + 0.033*slow' + 0.030*line' + 0.029*minute' + 0.023*hour' + 0.023*drive' + 0.022*car' + 0.020*food' + 0.019*thru'), (5, '0.076*food' + 0.054*terrible' + 0.051*lot' + 0.038*kid' + 0.036*area' + 0.030*star' + 0.028*neutral' + 0.028*parking' + 0.026*hot' + 0.026*big')}
```

```
topic_labels = {
    0: 'Bad service, wrong orders, rude staff',
    1: 'Fast and quick service, clean place, friendly staff',
    2: 'Overall good price, quality, service',
    3: 'Food quality issues (cold/old fries, burgers, nuggets)',
    4: 'Long wait times, slow drive-thru',
    5: 'Miscellaneous, general complaints about environment'
}
```

Bias & Uncertainty Validation

- Within the data we had to account for whether customers chose to leave a review (selection bias).
- We processed out unnecessary fields so that we would be left with reviews and ratings.
- Initially unsure of how to interpret the customer rating scale.
 - In order to combat some of the bias, we chose to interpret the ratings by treating 4–5 stars as positive, and 1–2 as negative, with 3 being neutral. This way, all of our analyses would use the same definition.
- LDA ignores word order. Choice of k and random seed affects the topics. We fixed the random seed in the LDA pipeline so that topic discovery and k-selection would be reproducible.



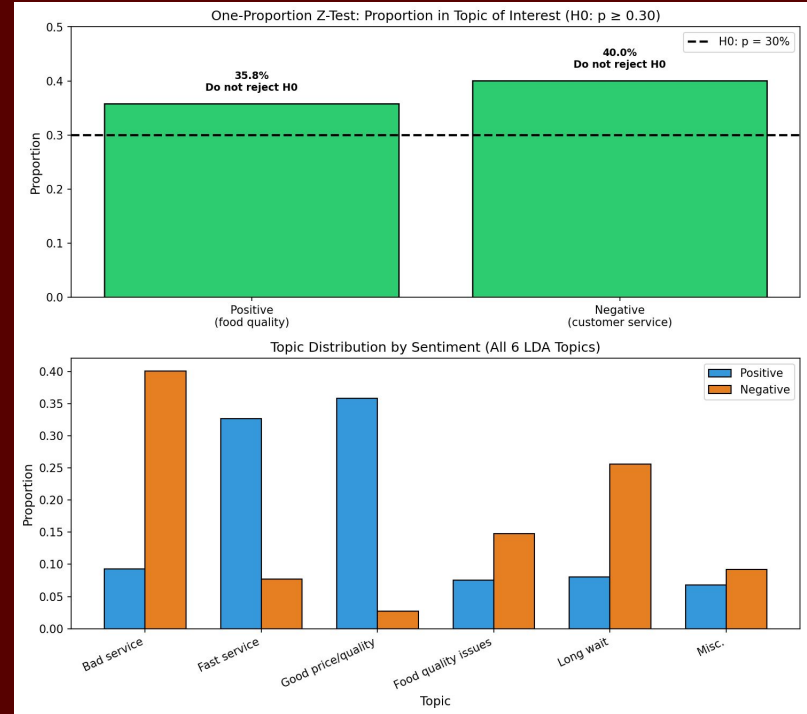
Results & Conclusions

Both hypothesis tests resulted in failing to reject the null hypothesis at the 0.05 significance level.

Food quality praises and customer service complaints occur at a rate consistent with our assumed benchmark.

Negative reviews are most heavily concentrated around bad service and long wait times, while positive reviews are more strongly associated with fast service and good price/quality.

Areas of improvement for McDonald's: Better service and faster wait times



Next Steps



New Lines of Exploration

Topic Variation by Geographic Location (different locations performance)

Mentions Over Time (rising prices, worsening customer service, etc.)



Project Improvements

Run LDA separately on positive and negative reviews.

Adjust our threshold based on more justification and reasoning.

More clearly defined topic groups to identify what constitutes good/bad food or service.



New Questions

How do different McDonald's locations compare to one another in terms of quality of food and customer service?

Is there a time of the year where McDonald's restaurants perform the worst?

References

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Thank you!

